

**THE MASTER QUADRATIC P_{G^*} AS A MATH–PHYSICS BRIDGE: χ_{-4} , THE
TERNARY SUBSTRATE, AND THE FTD IDENTIFICATION**

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ABSTRACT.

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1. SETTING AND SCOPE

We adopt the entire mathematical content of [1] without re-derivation. The constants in use are:

- $G^* = \Gamma(1/4)/\Gamma(3/4) \approx 2.9587$;
- $G_G = 1/\text{AGM}(1, \sqrt{2}) \approx 0.8346$, with the bridge identity $G^* = 2\sqrt{\pi} G_G$;
- $\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \{\pm 1\}$, the unique non-trivial Dirichlet character mod 4;
- the master quadratic $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$, with integer coefficient $16 = |\text{Aut}_{\mathbb{C}}(E_{\text{lemn}})|^2$ and roots $x_{\pm}$ satisfying Vieta's identities $x_+ + x_- = 16G^{*2}$, $x_+x_- = 16G^{*3}$, $1/x_+ + 1/x_- = 1/G^*$.

The mathematical paper [1] proves that χ_{-4} admits the explicit form $\chi_{-4}(n) = \text{Im}(i^n) = \sin(\pi n/2)$ and that its value set is

$$\{\chi_{-4}(n) : n \in \mathbb{Z}\} = \{i^2, 0, |i^2|\} = \{-1, 0, +1\}.$$

Numerical observation: the roots of P_{G^*} are $x_+ \approx 137.036\dots$ (matching the inverse fine-structure constant α^{-1} to 1.26 ppm) and $x_- \approx 3.024\dots$ (matching the number of QCD colours $N_c = 3$ to 0.8%). The Foundational Ternary Dynamics framework [2] posits the identification $(x_+, x_-) = (\alpha^{-1}, N_c)$ as a strongly motivated conjecture.

This note is the math-physics bridge: it draws structural consequences of the χ_{-4} -unification of [1] for the FTD identification, without proving the identification itself. All load-bearing claims are tagged [SYNTHESIS] or [STRONGLY MOTIVATED CONJECTURE] as appropriate; no new mathematical theorem is claimed in this note.

2. THE VALUE-SET COINCIDENCE AND THE GAUSSIAN-INTEGER-MODULE ONTOLOGY

Observation (Value-set coincidence). [Synthesis.] Three independent starting points terminate at the same three-element set:

- (1) the FTD physical primitive — each lattice site v takes a state $s_v \in \{-1, 0, +1\}$ encoding *negative state, void, positive state* [2];
- (2) the value set of the unique non-trivial real Dirichlet character of conductor 4 [1, §16];
- (3) the closure of the imaginary-unit orbit $\{i^n : n \in \mathbb{Z}\}$ under the imaginary-part map: $\{\text{Im}(i^0), \text{Im}(i^1), \text{Im}(i^2), \text{Im}(i^3)\} = \{0, 1, 0, -1\}$.

That these three sets coincide is a set-level identity, not yet a *functorial* identification. Whether the FTD voxel alphabet *arises from* the character χ_{-4} , or only happens to share its value set, is a physical-modelling question that lies outside the scope of the mathematical paper.

Remark (Candidate ontology: $\mathbb{Z}[i]$. -substrate realism) [Synthesis.] The value-set coincidence of Observation 2 suggests a working ontology in which the empirical universe is the real-coordinate readout of a $\mathbb{Z}[i]$ -module structure. Layering the established FTD framework against this ontology:

- voxels are positions in a $\mathbb{Z}[i]^d$ -style lattice ($d = 3$ in current FTD);
- voxel states $s_v \in \{-1, 0, +1\}$ are the values $\text{Im}(i^\bullet)$ readouts of the unit group $\mathbb{Z}[i]^\times$;
- the flux field $J \in \mathbb{R}^3$ is the continuous real part of an underlying complex object;
- time is the order parameter of the cyclic action $\langle i \rangle$ at long wavelength;
- the mass-length calibration $a_{\text{phys}} \equiv \ell_P$ is the choice of unit on the $\mathbb{Z}[i]$ -module;
- dimensionless physical constants (α , N_c , mass ratios) are intrinsic invariants of the $\mathbb{Z}[i]$ -module structure.

The ontological position is most accurately described as *Pythagorean structural realism, specialised to the Gaussian-integer module*: Pythagorean because number is taken as the substance from which the physical emerges; structural realist in attitude (entities are nodes in the structural network rather than independent objects — compare [4], though our specialisation to a single algebraic structure goes beyond their naturalistic stance); specialised because the particular structure is the

maximal order of $\mathbb{Q}(i)$ rather than an arbitrary mathematical structure. This is a more specific, and therefore more falsifiable, restriction of an MUH-style position [3]: not “all mathematical structures are physical” but “the physical is the $\mathbb{Z}[i]$ -module slice of the mathematical.” Under this specialisation Wigner’s “unreasonable effectiveness” [5] becomes a definitional consequence rather than a mystery.

Remark (Falsifiability). **[Synthesis.]** The $\mathbb{Z}[i]$ -substrate ontology is empirically refutable: if experiment ever produced an observable that requires a $\mathbb{Z}[\rho]$ -module structure (Eisenstein, sixfold) that cannot be embedded in a $\mathbb{Z}[i]$ ambient, or that requires $\mathbb{Z}[\sqrt{-2}]$ or $\mathbb{Z}[\sqrt{-7}]$ (other class-number-1 quadratic orders), the framework breaks. The position has therefore the falsifiability profile of a substantive empirical hypothesis, not a metaphysical claim.

3. THE BORN RULE AND χ_{-4} : A SCHEMA AND A CONJECTURE

The χ_{-4} -unification of [1] makes available a structural schema that connects two otherwise unrelated operations.

Observation (Born rule and χ_{-4} as two instances of a real-coordinate schema). **[Synthesis.]** The Born rule of quantum mechanics and the character χ_{-4} are two instances of a general schema “complex object $\rightarrow$ single real coordinate $\rightarrow \mathbb{R}$,” chosen independently in QM and in number theory for unrelated structural reasons:

- **Born rule:** $\psi \in \mathbb{C} \mapsto |\psi|^2 \in \mathbb{R}_{\geq 0}$, the squared modulus, a continuous map whose load-bearing content is the Gleason-theorem link between an inner-product-space amplitude and a probability measure.
- **Character χ_{-4} :** $i^n \in \langle i \rangle \subset \mathbb{C} \mapsto \text{Im}(i^n) \in \{-1, 0, +1\}$, the imaginary-part trace on a finite cyclic group, a discrete map whose content is the detection of quadratic residues mod 4.

The two operations differ in their domain (continuous Hilbert space vs. finite cyclic group), their codomain (positive half-line vs. signed three-point set), the coordinate they read (modulus vs. imaginary part), and their non-linearity (squared modulus vs. linear imaginary part). The observation worth recording is therefore not that they are “the same arrow,” but that both produce ternary or near-ternary outcomes when restricted to relevant subdomains, by virtue of the imaginary-unit orbit $\langle i \rangle$ having exactly three distinct imaginary projections.

Remark (Reflexive projection at two identical scopes and one analogous scope). **[Synthesis.]** Under the FTD framework’s identification of the voxel state field with the substrate’s reflexive projection [2], two scopes are *literally the same operation* and a third is *conjecturally analogous*:

- (Identical) The FTD voxel state field as the imaginary-part readout of $\mathbb{Z}[i]^\times \cup \{0\}$, and the character χ_{-4} as $\text{Im} \circ (i^\bullet)$ on the angular index, are literally the same map; their coincidence is a definitional consequence of the $\mathbb{Z}[i]$ -substrate position rather than an independent identification.
- (Analogous) The Born rule of quantum mechanics is conjecturally related to this same map at a derived scope (amplitude measurement of $\mathbb{Z}[i]$ -structured quantum states). This is the load-bearing physical conjecture, tagged **[STRONGLY MOTIVATED CONJECTURE]**; it is not established by any mathematical theorem and the differences enumerated in Observation 3 mean it should not be conflated with the two identical scopes.

4. THE MASTER QUADRATIC AS THE BRIDGE

The FTD bridge from χ_{-4} -derived mathematics to physical observables runs through the master quadratic P_{G^*} . We recall (from [1, Theorem 16.2]) that χ_{-4} generates the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection) and the constant G^* (Chowla–Selberg projection), but does *not* generate the specific polynomial form $x^2 - 16G^{*2}x + 16G^{*3}$ from classical CM/Weber/Hecke machinery (the three negative tests of [1, §16.5]).

The exponent pair $(2, 3)$ on G^* in the polynomial coefficients is nevertheless *uniquely picked out* by joint root-matching:

Observation (Joint-matching uniqueness, restated from [1, §16.6]). **[Synthesis.]** Among polynomial forms $y^2 - 16R^p y + 16R^q$ at $R = G^*$ with $(p, q) \in \mathbb{Z}^2$, $|p|, |q| \leq 5$, only the pair $(p, q) = (2, 3)$ produces an ordered root pair (y_+, y_-) that simultaneously matches the FTD pair (α^{-1}, N_c) to sub-percent precision. The match precision at $(2, 3)$ is 1.26 ppm for $y_+ = 137.036$ and 0.8% for $y_- = 3.024$. Other exponent pairs match one root but not both, or neither.

Remark (Information-theoretic minimality of the FTD bridge). **[Synthesis.]** The FTD math-physics bridge requires exactly *a single discrete integer-pair selector* from the physical side — the exponent-pair selector $(p, q) \in \{(2, 3), (3, 4), (4, 5), \dots, (1, 2), (2, 4), \dots\}$ — and that bit is over-determined by joint matching to two distinct physical constants (α^{-1}, N_c) . Equivalently: the FTD master quadratic is the unique polynomial of the form $y^2 - 16R^p y + 16R^q$ at $R = G^*$ whose ordered root pair (y_+, y_-) jointly matches (α^{-1}, N_c) to sub-percent precision.

This is not a derivation of the bridge — the joint match itself is the input that selects (p, q) — but it is the cleanest available *minimality statement*: the math-physics bridge in the FTD framework requires no parameters beyond the choice of one exponent pair, and the choice is over-determined (hence robust) against the joint match to two distinct physical constants.

Observation (The locus of the math-physics gap). Combining [1, §16.5] (no CM-internal source for $(2, 3)$) with Observation 4 (joint match uniquely fixes $(2, 3)$): the gap between χ_{-4} -derived mathematical structure and the FTD physical identification is localised precisely at the polynomial *assembly* step. The integer coefficient 16 and the constant G^* are both forced by χ_{-4} . The exponent pair $(2, 3)$ is forced only by the joint physical-matching requirement. Closing this last gap — if a clean derivation of $(2, 3)$ from χ_{-4} -internal data exists — would convert the FTD bridge from a two-arrow conjecture into a single-arrow theorem.

5. THE FULL DESCENT CHAIN

The cumulative content of [1] and this companion can be displayed as a single descent chain from physics to ontological bedrock:

$$\begin{array}{c}
 \underbrace{(\alpha^{-1}, N_c)}_{\text{measured physical constants}} \\
 \uparrow \text{ [STRONGLY MOTIVATED CONJECTURE] (FTD bridge)} \\
 \underbrace{P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}}_{\text{master quadratic}} \\
 \uparrow (2, 3) \text{ exponents: uniquely forced by joint root match [1, §16.6]} \\
 \underbrace{\{16, G^*\}}_{\text{building blocks}} \\
 \uparrow \text{ four functorial projections [1, Thm 16.2]} \\
 \underbrace{\chi_{-4}}_{\text{Kronecker character of } \mathbb{Q}(i)} \\
 \uparrow \text{ imaginary-part trace } \chi_{-4}(n) = \text{Im}(i^n) = \sin(\pi n/2) \\
 \underbrace{\{-1, 0, +1\}}_{\text{value set = FTD voxel alphabet (set-level)}} \\
 \uparrow \text{ closure of } \langle i \rangle\text{-orbit under squaring \& modulus} \\
 \underbrace{i^2 = -1}_{\text{ontological zero-point}}
 \end{array}$$

Every arrow above the FTD bridge is established (theorem, observation, or honest conjecture) in [1]. The FTD bridge itself is the physical conjecture that this companion records under [STRONGLY MOTIVATED CONJECTURE]. Everything below the bridge is pure mathematics; the only place that requires physical input is the single bit of the exponent-pair selector.

6. OUTLOOK

Two productive lines of work are identified by the descent chain.

Closing the bridge. If a derivation of the exponent pair (2, 3) from χ_{-4} -internal data can be made rigorous — the candidate framing in [1, §16.5, Theorem (Sym²–Sym³ uniqueness, leading-period)] is the most natural starting point — the FTD bridge would become a single-arrow theorem rather than a two-arrow conjecture. This would not prove the empirical match to (α^{-1}, N_c) (that remains an experimental fact about the universe), but it would localise the conjectural content to a single physical assertion: that the universe’s electroweak and strong sectors are built from a $\mathbb{Z}[i]$ -substrate.

Extending the bridge. The current FTD identifications cover α and N_c . The Standard Model has additional dimensionless constants ($\sin^2 \theta_W$, m_μ/m_e , m_τ/m_e , PMNS angles, etc.) which the FTD framework currently parameterises without first-principles derivation. The χ_{-4} -unification suggests that further dimensionless invariants may be discoverable as roots of higher-rank master polynomials living in $\text{Sym}^\bullet(H^1(E_{\text{lemn}}))$, or as Hecke L-values of related CM elliptic curves. This is the natural program for extending the bridge from (α^{-1}, N_c) to the full SM.

A second-tier programme suggested by the parallel equianharmonic structure of [1, §16.4] is the identification of the χ_{-3} -side (equianharmonic, $|\text{Aut}| = 6$) as the structural home of the QCD gauge structure, with χ_{-4} (lemniscatic) as the home of the electroweak structure. The Moore Layer Theorem of FTD [2] produces $U(1) \times SU(2) \times SU(3)$ from a 27-block polyhedral decomposition;

whether this can be reformulated as the joint action of χ_{-4} (giving $U(1)$ and part of $SU(2)$) and χ_{-3} (giving $SU(3)$ and the rest of $SU(2)$ via the Hurwitz-quaternion intersection $\mathbb{H} \supset \mathbb{Z}[i], \mathbb{Z}[\rho]$) is an open speculative direction worth investigating but not pursued here.

REFERENCES

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